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Practical 8

In [5]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = sns.load_dataset('titanic')
data.head()
```

Out[5]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

In [6]:
```
data.describe()
```

Out[6]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [7]:

Out[7]:

```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

In [8]:
```
data.info
```

Out[8]:
```
<bound method DataFrame.info of      survived  pclass     sex   age  sibsp  parch     fare embarked   class  \
0           0       3    male  22.0      1      0   7.2500        S   Third
1           1       1  female  38.0      1      0  71.2833        C   First
2           1       3  female  26.0      0      0   7.9250        S   Third
3           1       1  female  35.0      1      0  53.1000        S   First
4           0       3    male  35.0      0      0   8.0500        S   Third
..        ...     ...     ...   ...    ...    ...      ...      ...     ...
886         0       2    male  27.0      0      0  13.0000        S  Second
887         1       1  female  19.0      0      0  30.0000        S   First
888         0       3  female   NaN      1      2  23.4500        S   Third
889         1       1    male  26.0      0      0  30.0000        C   First
890         0       3    male  32.0      0      0   7.7500        Q   Third

       who  adult_male deck  embark_town alive  alone
0      man        True  NaN  Southampton    nc  False
1    woman       False    C    Cherbourg   yes  False
2    woman       False  NaN  Southampton   yes   True
3    woman       False    C  Southampton   yes  False
4      man        True  NaN  Southampton    nc   True
..     ...         ...  ...          ...   ...    ...
886    man        True  NaN  Southampton    nc   True
887  woman       False    B  Southampton   yes   True
888  woman       False  NaN  Southampton    nc  False
889    man        True    C    Cherbourg   yes   True
890    man        True  NaN   Queenstown    nc   True

[891 rows x 15 columns]>
```

In [12]:
```
data['age'] = data['age'].fillna(np.mean(data['age']))
data['embark_town'] = data['embark_town'].fillna(data['embark_town'].mode()[0])
data['embarked'] = data['embarked'].fillna(data['embarked'].mode()[0])
```

In [14]:
```
data.drop(columns=['deck'], inplace=True)
```

In [15]:
```
data.isnull().sum()
```

Out[15]:
```
survived       0
pclass         0
sex            0
age            0
sibsp          0
parch          0
fare           0
embarked       0
class          0
who            0
adult_male     0
embark_town    0
alive          0
alone          0
dtype: int64
```

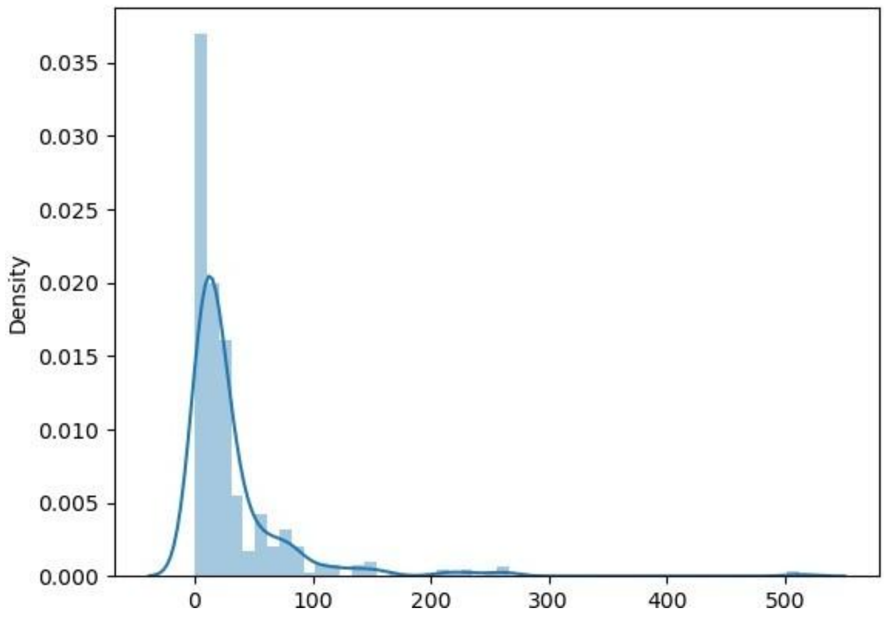
In [16]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 14 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  ------
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          891 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     891 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  embark_town  891 non-null    object
 12  alive        891 non-null    object
 13  alone        891 non-null    bool
dtypes: bool(2), category(1), float64(2), int64(4), object(5)
memory usage: 79.4+ KB
```

In [21]:
```
sns.distplot(x = data['fare'])
```

```
C:\Users\GAYATRI\AppData\Local\Temp\ipykernel_12140\4107250924.py:1:  UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(x = data['fare'])
```

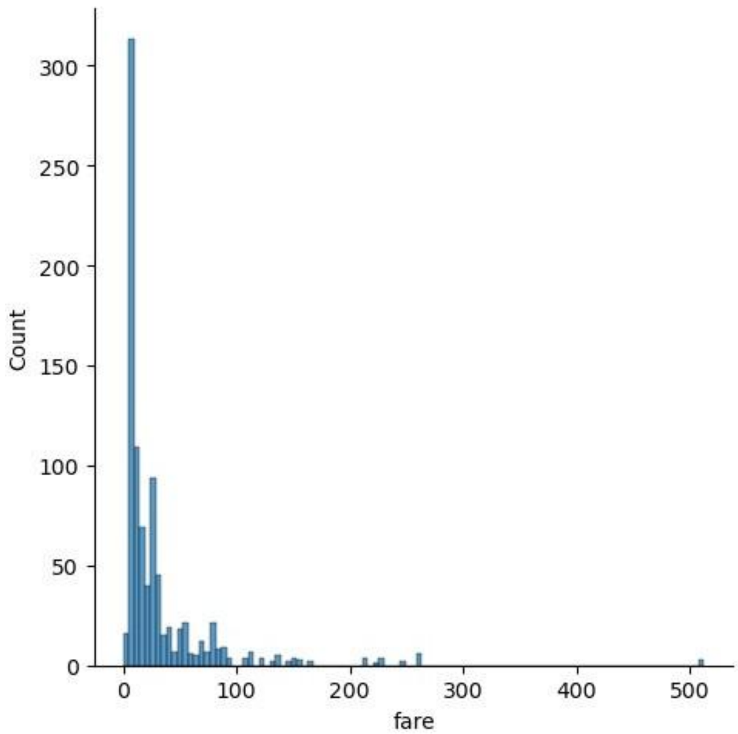
Out[21]:
```
<Axes: ylabel='Density'>
```



In [23]:
```
sns.displot(x = data['fare'])
```

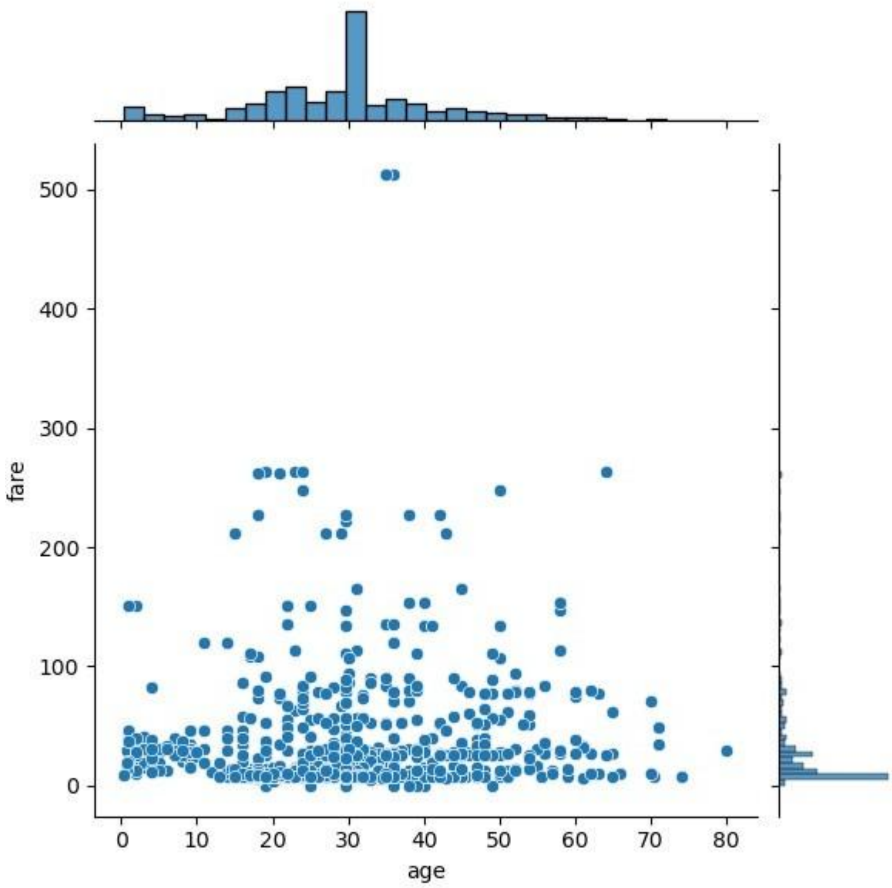
```
C:\Users\GAYATRI\anaconda3\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has change
d to tight
  self._figure.tight_layout(*args, **kwargs)
```

Out[23]:
```
<seaborn.axisgrid.FacetGrid at 0x24e46097610>
```



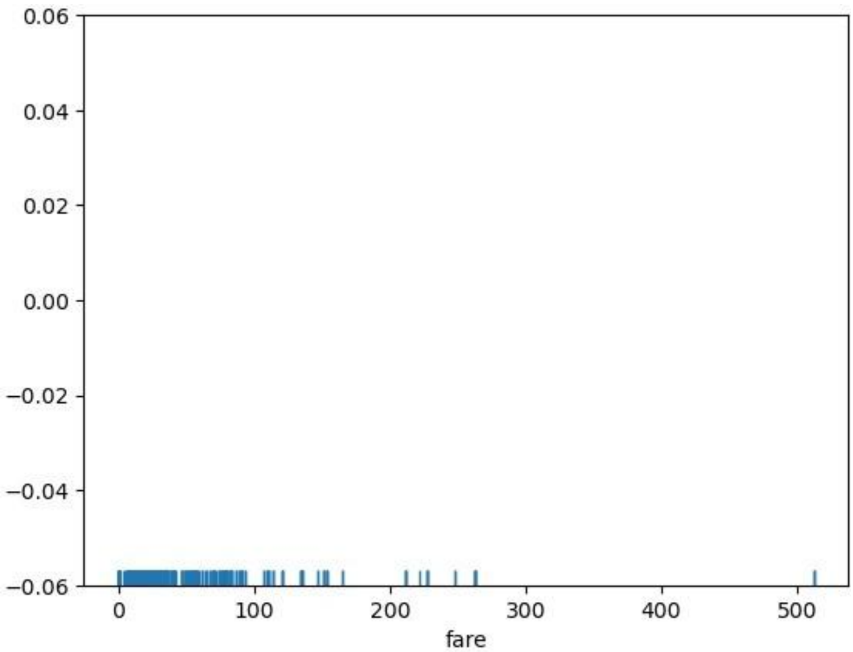
In [24]:
```
sns.jointplot(x = data['age'], y = data['fare'], kind = 'scatter')
```

Out[24]:
```
<seaborn.axisgrid.JointGrid at 0x24e45f50450>
```



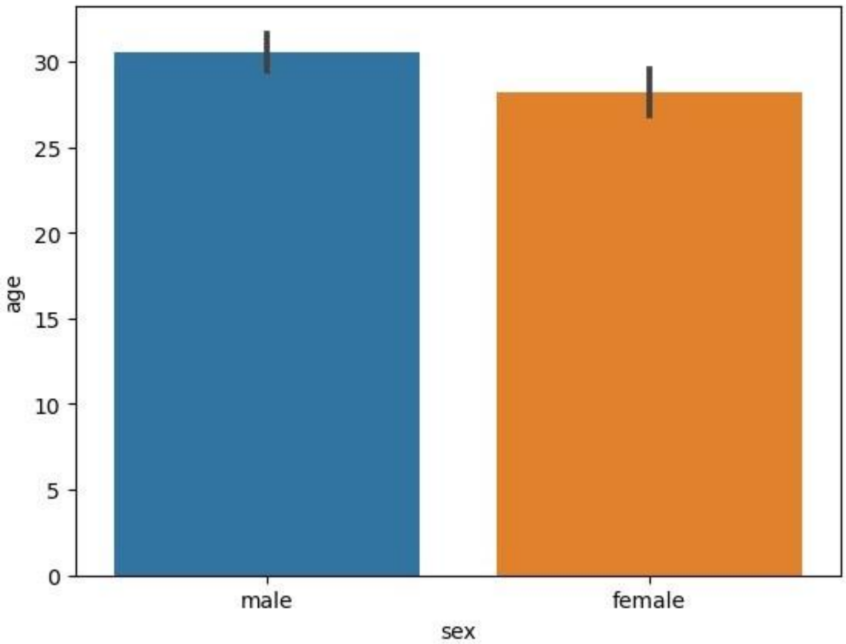
In [25]:
```
sns.rugplot(dataset['fare'])
```

Out[25]:
```
<Axes: xlabel='fare'>
```



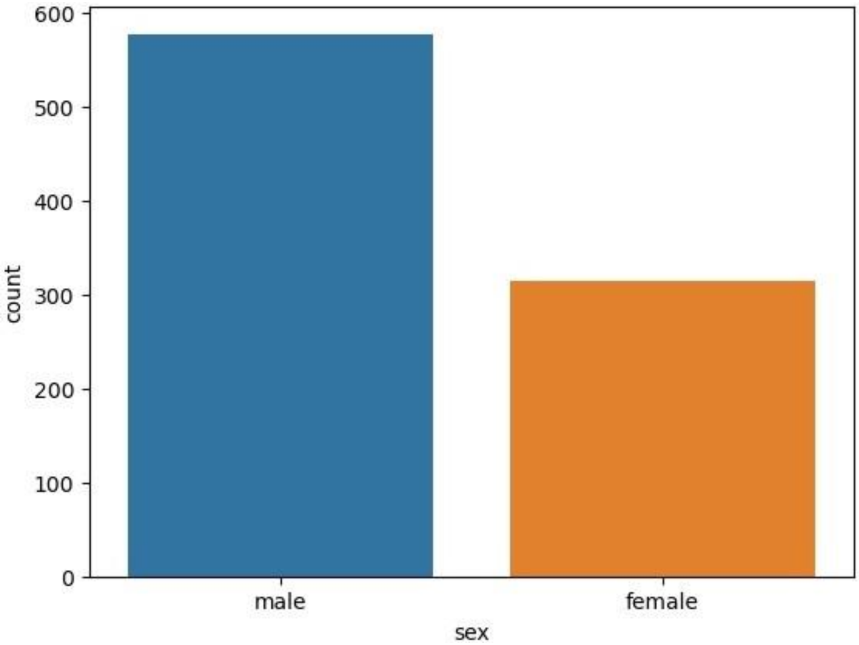
In [26]:
```
sns.barplot(x='sex', y='age', data=data)
```

Out[26]:
```
<Axes: xlabel='sex', ylabel='age'>
```



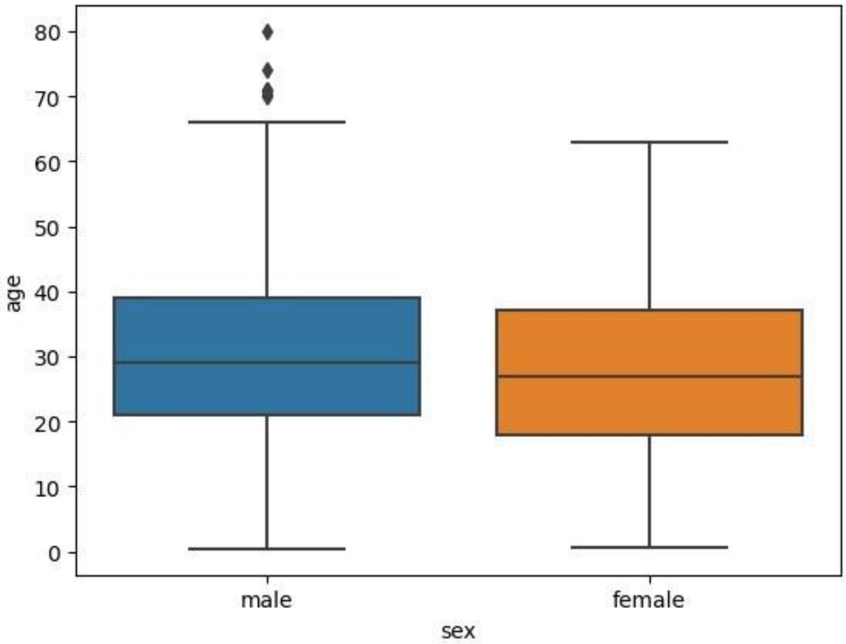
In [27]:
```
sns.countplot(x='sex', data=dataset)
```

Out[27]: <Axes: xlabel='sex', ylabel='count'>



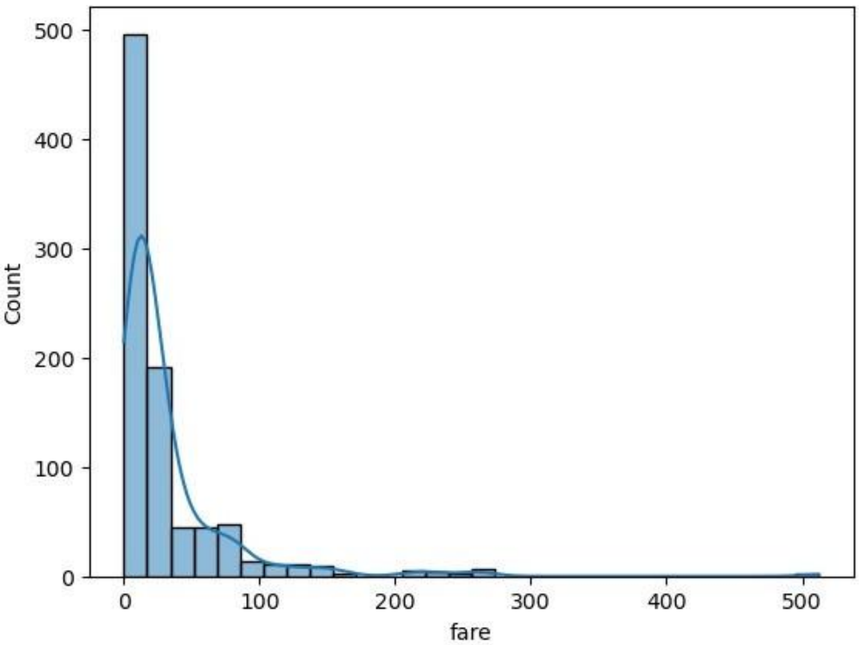
In [28]:
```
sns.boxplot(x='sex', y='age', data=dataset)
```

Out[28]: <Axes: xlabel='sex', ylabel='age'>



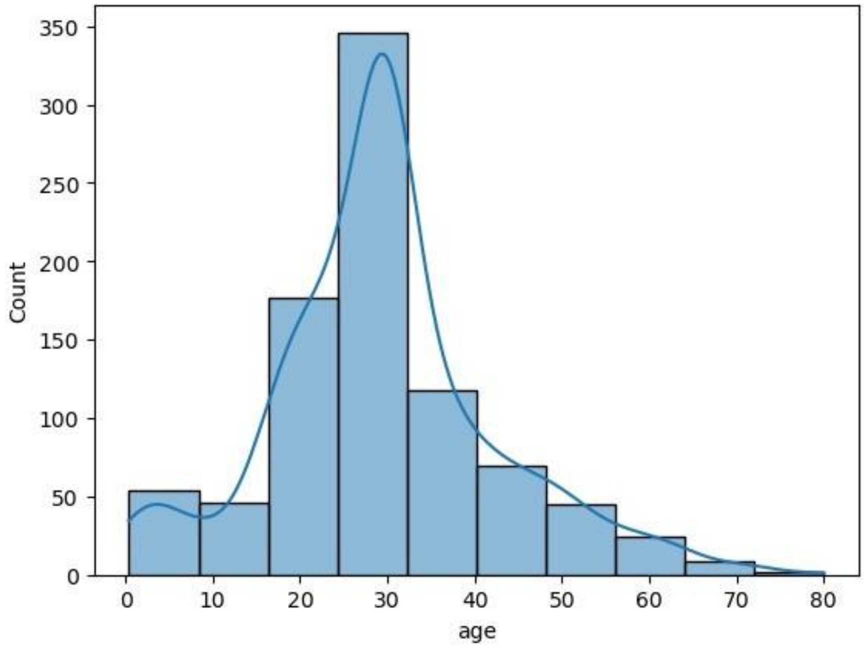
In [37]:
```
sns.histplot(data=data, x='fare', bins=30, kde=True)
```

Out[37]: <Axes: xlabel='fare', ylabel='Count'>



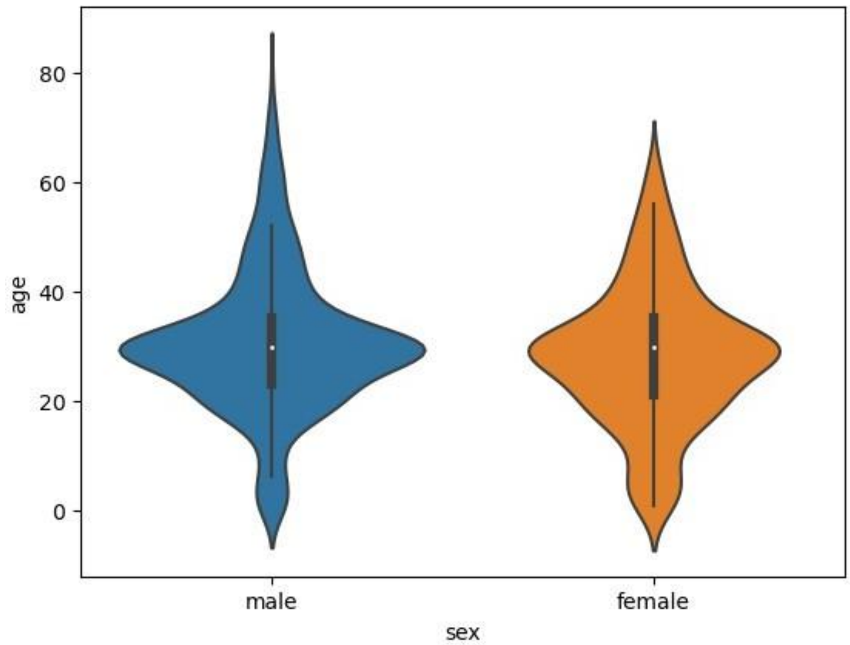
In [38]:
```
sns.histplot( data=data,x='age', bins=10, kde=True)
```

Out[38]: <Axes: xlabel='age', ylabel='Count'>



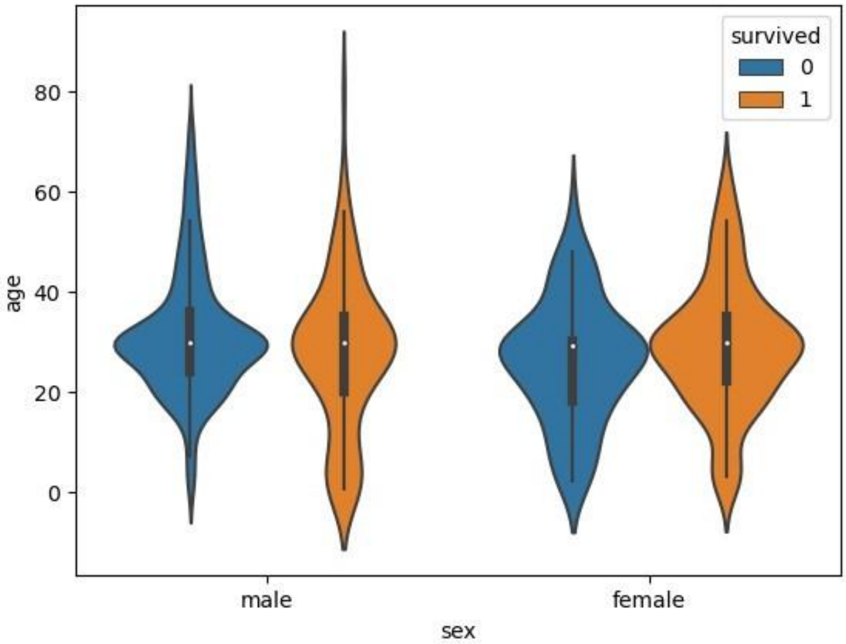
In [39]:
```
sns.violinplot(x='sex', y='age', data=data)
```

Out[39]: <Axes: xlabel='sex', ylabel='age'>



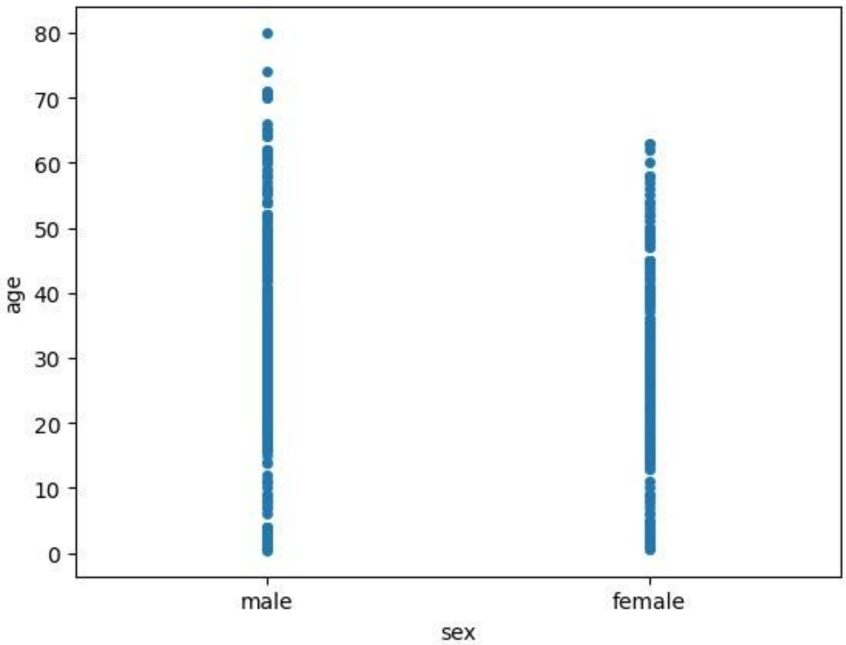
In [40]:
```
sns.violinplot(x='sex', y='age', data=data, hue='survived')
```

Out[40]: <Axes: xlabel='sex', ylabel='age'>



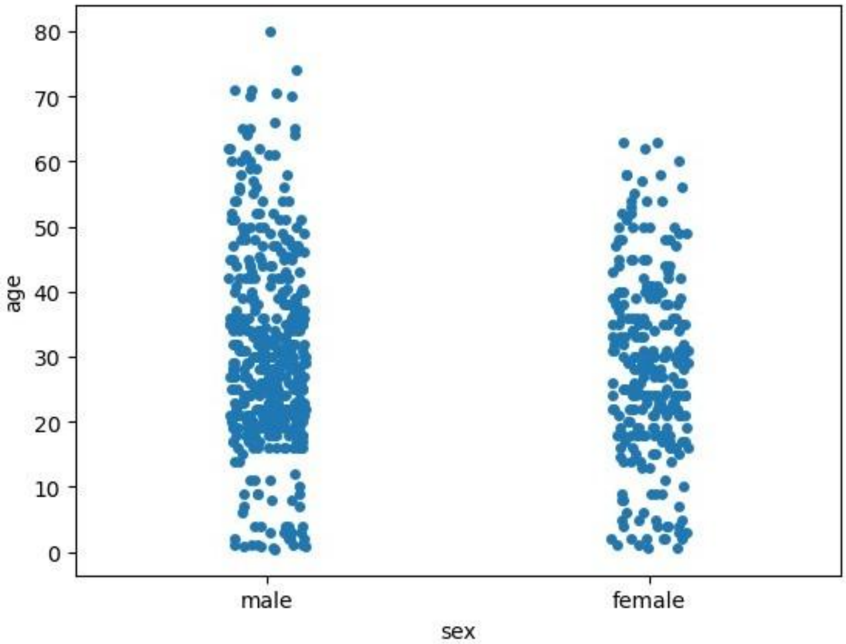

```
In [41]: sns.stripplot(x='sex', y='age', data=dataset, jitter=False)
```

Out[41]: <Axes: xlabel='sex', ylabel='age'>




```
In [42]: sns.stripplot(x='sex', y='age', data=dataset, jitter=True)
```

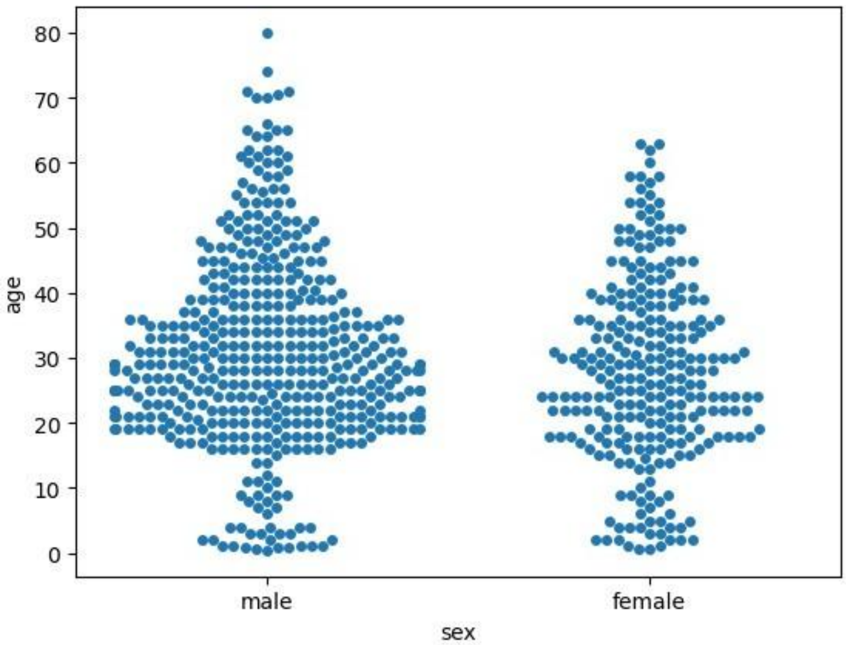
Out[42]: <Axes: xlabel='sex', ylabel='age'>



In [43]:

```
sns.swarmplot(x='sex', y='age', data=dataset)
```

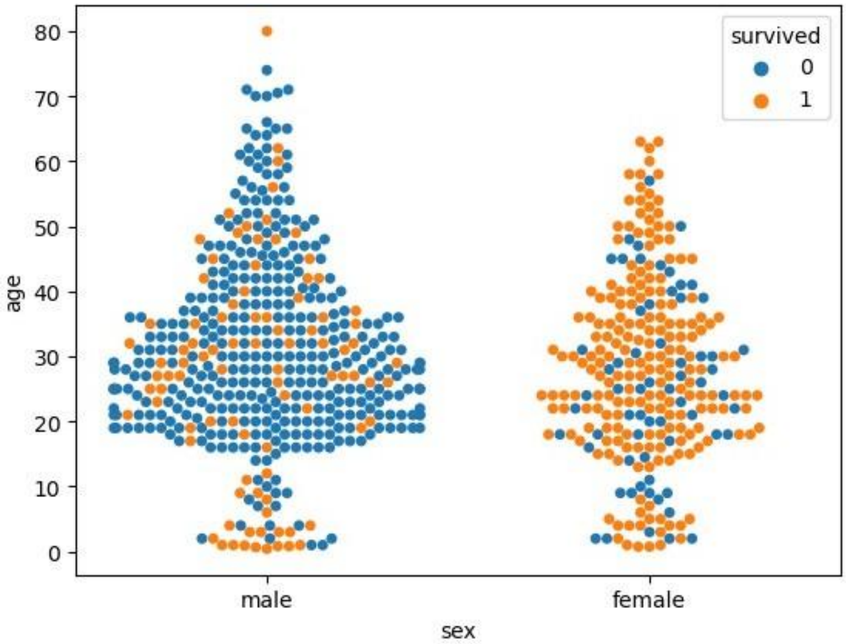
Out[43]: <Axes: xlabel='sex', ylabel='age'>



In [44]:

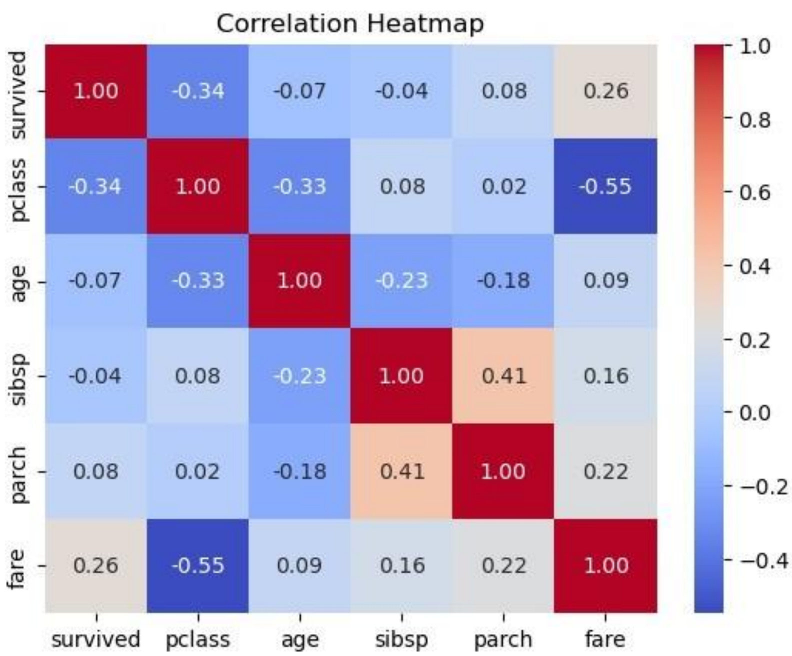
```
sns.swarmplot(x='sex', y='age', data=dataset, hue='survived')
```

Out[44]: <Axes: xlabel='sex', ylabel='age'>



In [49]:

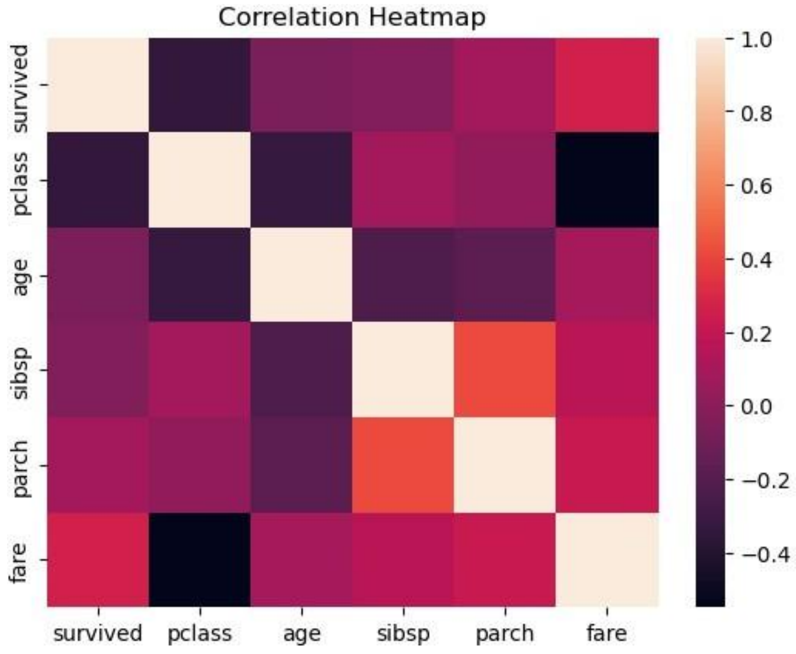
```
import seaborn as sns
import matplotlib.pyplot as plt

# Select only numeric columns for correlation
numeric_data = data.select_dtypes(include='number')

# Calculate correlation
corr = numeric_data.corr()

# Plot heatmap
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
plt.show()
```



In [52]:

```
sns.heatmap(corr)
plt.title('Correlation Heatmap')
```

```
plt.show()
```


Correlation Heatmap
survived
pclass
age
sibsp
parch
fare
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

In [ ]: